

KASHIF MUNEEB

AI / ML Engineer

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Education

University of Engineering and Technology, Lahore

Bachelor of Science in Computer Engineering • CGPA: 3.12/4.00

Nov. 2021 – Aug 2025

Lahore, Punjab, Pakistan

The KIMS College, Chiniot

Intermediate in Computer Science (ICS) • Marks: 1085/1100

Aug. 2019 – July 2021

Chiniot, Punjab, Pakistan

Experience

AI Engineer Intern

Dec 2025 – Present

AI4LYF

Remote

- Designed and validated a cough-based biometric verification pipeline for per-user identity authentication from cough acoustics: enrollment-and-verification workflow with real-world blind-test evaluation; achieved **1.7% EER** and **94.6% verification accuracy** on a held-out test set — outperforming published reference baselines on standard biometric metrics.
- Integrated the personalized verification stage with the existing on-device cough detection pipeline into a unified two-stage acoustic processing system; modular stage handoff with independent deployability, enabling cough-type classification and per-user identity verification from a single audio stream.
- Built an internal-testing web app for the verification workflow: enrollment and verification from consumer-hardware microphones, automated blind-test runner, per-session Excel result logging, and microphone device selection — used internally by the team for blind-test evaluation.
- Driving on-device readiness for the verification stage toward iOS handoff: mobile-format model export, inference-path profiling, and footprint alignment with the production classification system — supporting personalization integration with the iOS dev team through model handoff and on-device validation.

Generative AI Intern

June 2024 – August 2024

Xavor Corporation

Lahore, Pakistan

- Implemented NLP preprocessing (stop-word removal, stemming, lemmatization) to improve text data quality and downstream ML model performance.
- Developed and evaluated feature extraction methods (BoW, TF-IDF, n-grams) and word embeddings (Word2Vec, GloVe) for semantic representation in NLP tasks.
- Built Retrieval-Augmented Generation (RAG) pipelines using Pinecone, ChromaDB, Hugging Face, and OpenAI embeddings to enhance contextual response accuracy.
- Fine-tuned LLMs (Llama 2, Gemma) using LoRA/QLoRA and deployed UIs with Streamlit.

Final Year Project

Smart Fire Extinguisher System | Machine Learning, Computer Vision

June 2024 – May 2025

- Developed a CV-based system using deep learning for early fire detection with a smart camera sensor.
- Implemented real-time smoke and fire recognition leveraging YOLO-based object detection and image processing.
- Integrated an automated fire suppression with a motor pump system that dynamically adjusts water spray trajectory.
- Designed a fire monitoring module that maps locations using coordinate-based calculations for precise response.

Projects

Pakistan Law Assistant | Python, RAG, LangChain, FAISS, Streamlit

March 2026

- Built a Retrieval-Augmented Generation (RAG) assistant for legal question-answering grounded in Pakistani legal documents, combining dense vector search with keyword-based retrieval (hybrid FAISS + BM25) for higher accuracy than single-retriever baselines.
- Integrated multilingual embeddings for English/Urdu queries with a Groq-hosted Llama 3.3 70B backend; deployed publicly on Streamlit Cloud.

Multi-Agent Research Agent | Python, LangGraph, Groq API

March 2026

- Designed a 7-agent research pipeline using LangGraph that takes a user question and returns a structured, cited research report — covering query decomposition, web search, source evaluation, and synthesis.
- Integrated Groq-hosted Llama 3.3 70B for agent reasoning and DuckDuckGo for live web retrieval; tuned query specificity to surface benchmark data over generic definition pages.

- Built an NLP-powered review analysis dashboard that scrapes competitor app reviews from Google Play and the App Store, clusters them by topic using sentence embeddings, and surfaces top user pain points and praise patterns.
- Used all-MiniLM-L6-v2 embeddings with K-Means clustering and automatic k selection via silhouette score; added per-cluster TF-IDF keyword extraction and star-rating-based sentiment analysis.

Ghar Ka Hisaab (Household Expense Tracker) | *React Native, Android, Google Apps Script, Groq API* Feb – April 2026

- Built a cross-platform expense-tracking app (Android + web) using Expo/React Native with a Google Apps Script backend and Netlify-hosted web frontend.
- Integrated Groq-hosted Llama 3.3 70B to parse free-form expense messages into structured entries — users type natural language (e.g., “Ali paid 2000 for groceries”) and the LLM extracts amount, payer, and category automatically.

Technical Skills

Languages: Python, SQL, JavaScript, HTML/CSS

Deep Learning: TensorFlow/Keras, PyTorch, 1D/2D CNNs, ResNets, transfer learning, triplet/metric learning, model evaluation (EER, GAR/FAR, d-prime)

Machine Learning: Scikit-learn, classification & regression, feature engineering, cross-validation, blind-test protocols

LLMs & NLP: Hugging Face Transformers, LangChain, LangGraph, LoRA/QLoRA fine-tuning, prompt engineering, RAG pipelines (FAISS, Pinecone, ChromaDB, BM25 hybrid retrieval)

Audio & Signal Processing: Librosa, SoundDevice, mel spectrograms, peak-based segmentation, loudness normalization (BS.1770), real-time microphone capture

Computer Vision: OpenCV, YOLO-based object detection, image preprocessing

Data & Backend: Pandas, NumPy, Matplotlib, openpyxl, FAISS, SQL

Deployment & Tooling: Streamlit, Docker, Conda, Git/GitHub, Jupyter, VS Code, PyCharm, Raspberry Pi, Groq API, Azure AI

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

March 2026

Microsoft | Credential ID: 48634ADF1E5F3B3

Online

Artificial Intelligence and Machine Learning

Sep 2025 – Nov 2025

NAV TTC — Prime Minister Youth Skill Development Program

ShellShift, Lahore

Generative AI Bootcamp Internship

August 2024

Xavor Corporation

Lahore, Pakistan

Artificial Intelligence and Machine Learning

August 2024

ACM Fellowship Program 2024

Online

Introduction to Relational Databases (RDBMS)

January 2024

Coursera — Authorized by IBM

Online

Introduction to Web Development with HTML, CSS, JavaScript

June 2024

Coursera — Authorized by IBM

Online